Sam McClenaghan

780-221-1327 | sam@aream.ca | smccl.ca

Education

University of Victoria — Victoria, BC
Honours Bachelor of Science in Computer Science — *Sep 2023 - Dec 2027*
Computer Science, Minor in Mathematics | GPA: 7.8/9.0

Experience

Software Engineer, Intern — May 2026 - Present
Vertical City — *Victoria, BC*

- Reduced media processing latency from 5 minutes to seconds by implementing event-driven, in-process rendering for near-real-time display updates
- Built an AWS media pipeline with S3 and CloudFront serving 2M+ requests/day across 10,000+ screens, eliminating $20K in annual vendor costs
- Reduced bandwidth usage by 30% by developing a perceptual hashing system that identifies duplicate media across different encodings

Full-Stack Developer, Intern — May 2025 - Dec 2025
Leanpub — *Remote (Victoria, BC)*

- Implemented an AWS Lambda and Step Functions pipeline to generate image alt text at scale, improving accessibility coverage for 50,000+ ebooks
- Shipped a production rich text editor for 50,000+ authors by integrating Lexical with React and TypeScript into core content workflows

Software Developer, Intern — Nov 2022 - Aug 2024
Trust Science — *Edmonton, AB*

- Reduced AWS S3 storage costs by 15% by implementing a file version-control system with HTTP/2 and JSON Patch for efficient delta storage
- Increased test coverage from 35% to 80% with Playwright E2E tests and Jest unit tests

Projects

CourseFlow | *TypeScript, Cloudflare Workers, Durable Objects, D1* | courseflow.smccl.ca — 2025 - 2026

- Built and launched a course planner used by 2,000+ UVic students to create 3,000+ shareable timetables
- Implemented real-time schedule sharing using WebSockets and Cloudflare Durable Objects, broadcasting schedule changes and access revocations to connected viewers
- Developed a course recommendation engine combining embedding-based cosine similarity, BM25 lexical search, and course-level ranking, precomputing results in D1 to avoid request-time inference

Anchor | *TypeScript, PostgreSQL, Supabase, Vercel* — 2025

- Owned backend and deployment for a hackathon mental-health companion: PostgreSQL migrations, Garmin HRV data ingestion, and Vercel deployment

Technical Skills

Languages: C, Go, Python, Ruby, TypeScript, Java, JavaScript, SQL
Development & Testing: Docker, ffmpeg, RSpec, Jest, Playwright, Ruby on Rails, React
Cloud: AWS (S3, Lambda, Step Functions, CloudFront, DynamoDB, MediaConvert), Cloudflare Workers
Databases: MySQL, PostgreSQL, Redis, Elasticsearch